Brainstorming & Idea Prioritization

Brainstorming & Ideation Report - Rising Waters

1. Problem Statement

Floods represent one of the most destructive natural disasters worldwide, leading to loss of lives, mass displacement, and immense economic damage. Conventional weather forecasts are often too broad and do not provide localized early-warning details. Disaster management teams and local authorities lack tools to classify real-time flood susceptibility based on local meteorological parameters.

The Mission of Rising Waters: To build an end-to-end Machine Learning warning system that consumes real-time local weather parameters (Annual Rainfall, Cloud Cover, and seasonal distribution) to classify flood risks instantly, allowing disaster relief coordinators to plan evacuations and allocate resources proactively.

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2. Empathy Mapping

To understand the core users-such as local meteorologists, disaster relief officers, and community leaders-we analyzed their experiences:

Says (User quotes):

- "We need a faster way to convert rainfall readings into actionable evacuation plans."
- "The warning bulletins are often too late; by the time the alert goes out, streets are already waterlogged."
- "Can we get a specific safety confidence score instead of just a generic alert?"

Thinks (Beliefs & worries):

- *How do I prioritize resource allocation between three different sub-districts during monsoons?*
- *I worry that false alarms will make the community ignore actual evacuations.*
- *Is there a reliable model that generalizes weather trends without massive computational requirements?*

Does (Observed behaviors):

- Regularly monitors stream gauges and manually cross-references them with radio weather alerts.
- Uses spreadsheets to log monthly rainfall but struggles to predict peak hazards.
- Mobilizes volunteer teams on stand-by during active monsoon season based on gut feeling.

Feels (Emotions):

- Anxious about sudden cloudbursts during the night.
- Helpless due to the lack of specialized prediction tools for localized districts.
- Responsible for safeguarding lives and property under extreme conditions.

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3. Problem-Solution Fit

User Pain Point	System Feature	Strategic Value
Delayed warnings result in trapped	Real-time classification with custo	Evacuation plans can be initiated h

Inaccurate resource allocation.	Standardized parameter tables summa	Relief coordinators deploy sandbags
Complex, heavy software configurati	Zero-dependency Flask API compiled	Runs instantly in browser, compatib